

Natural Language Processing

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Text analysis Theory

Syntactic Analysis [POS tagging]

understand or/and reconstruct natural language data.

understand refers to knowing and identifying the grammatical structure of the sentence/token.

Semantic Analysis [NER]

goal: is to identify the context of the sentence/data.

stone.

rock emerald kidney stone

Natural Language Computing Science

NLP for feature generation

NLU

Text feature Handling for Inferential Statistics

Feature Transformation

→ goal is to make the text feature set compatible for the model algo/ NN.

Use-case: SMS spam classification

Given dataset.

(raw string) sms	label
Welcome to my world	s
NLP is interesting	h
NLU is awesome	h
Welcome Welcome	s

Step 1: Extract the vocabulary of the corpus.

a single record in text feature set.

DOCUMENT

→ Welcome to my world

→ NLP is interesting

→ NLU is awesome

→ Welcome Welcome

CORPUS

set of documents in a dataset

Word
Tokenization

['Welcome', to, my, world, NLP, is, interesting, NLU, is, awesome, Welcome, Welcome]

Extract unique words

[Welcome, to, my, world, NLP, is, awesome, NLU, interesting]

Remove stopwords

[Welcome, world, NLP, interesting, NLU, awesome]

VOCABULARY

they are unique words in a corpus without stopwords.

Step 2: Create BOW [Bag of words] / Document Term Matrix

The BOW model represents text as a collection of words, disregarding the grammar and word order but keeping the multiplicity of the word.

BOW is derived from the corpus and vocabulary and contains the statistical freq. of corpus.

		VOCABULARY					
DOCUMENT		welcome	world	NLP	interesting	NW	awesome
	Welcome to my world	1	1	0	0	0	0
	NLP is interesting	0	0	1	1	0	0
	NW is awesome	0	0	0	0	1	1
	Welcome Welcome	2	0	0	0	0	0

Step 3: Perform TF-IDF transformation.

TF → Term Frequency

It measures how frequently a term appears in a document.
(word) (sentence/paragraph/word)

Formula

$$TF(t, d) = \frac{\text{Number of times the term 't' appears in document 'd'}}{\text{Total no. of terms in document 'd'}}$$

IDF → Inverse Document Frequency.

It measures how important a term is across multiple documents.

Formula

$$IDF(t, D) = \log \left(\frac{\text{Total no. of documents}}{\text{Number of documents containing term 't'}} \right)$$

So TF-IDF looks like below

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) * \text{IDF}(t, D)$$

multiplication

$t \rightarrow$ term/word

$d \rightarrow$ individual document/record

$D \rightarrow$ corpus (set of documents)

eg: To calc the TF

	welcome	world	NLP	interesting	NLU	awesome
Document - Welcome to my world	0.25	0.25	0	0	0	0
NLP is interesting	0	0	0.33	0.33	0	0
NLU is awesome	0	0	0	0	0.33	0.33
Welcome Welcome	1	0	0	0	0	0

$$\text{TF}(\text{'welcome'}, \text{'Welcome to my world'}) = \frac{1}{4} = 0.25$$

To calc IDF,

$$\text{welcome} \rightarrow \log\left(\frac{4}{2}\right) = 0.69$$

$$\text{world} \rightarrow \log\left(\frac{4}{1}\right) = 1.39$$

$$\text{NLP} \rightarrow \log\left(\frac{4}{1}\right) = 1.39$$

$$\text{NLU} = \log\left(\frac{4}{1}\right) = 1.39$$

$$\text{interesting} = \log\left(\frac{4}{1}\right) = 1.39$$

$$\text{awesome} = \log\left(\frac{4}{1}\right) = 1.39$$

∴ TF-IDF matrix

document

	welcome	world	NLP	interesting	NLU	awesome
Welcome to my world	0.25×0.69	0.25×1.39	0	0	0	0
NLP is interesting	0	0	0.33×1.39	0.33×1.39	0	0
NLU is awesome	0	0	0	0	0.33×1.39	0.33×1.39
Welcome Welcome		0	0	0	0	0

